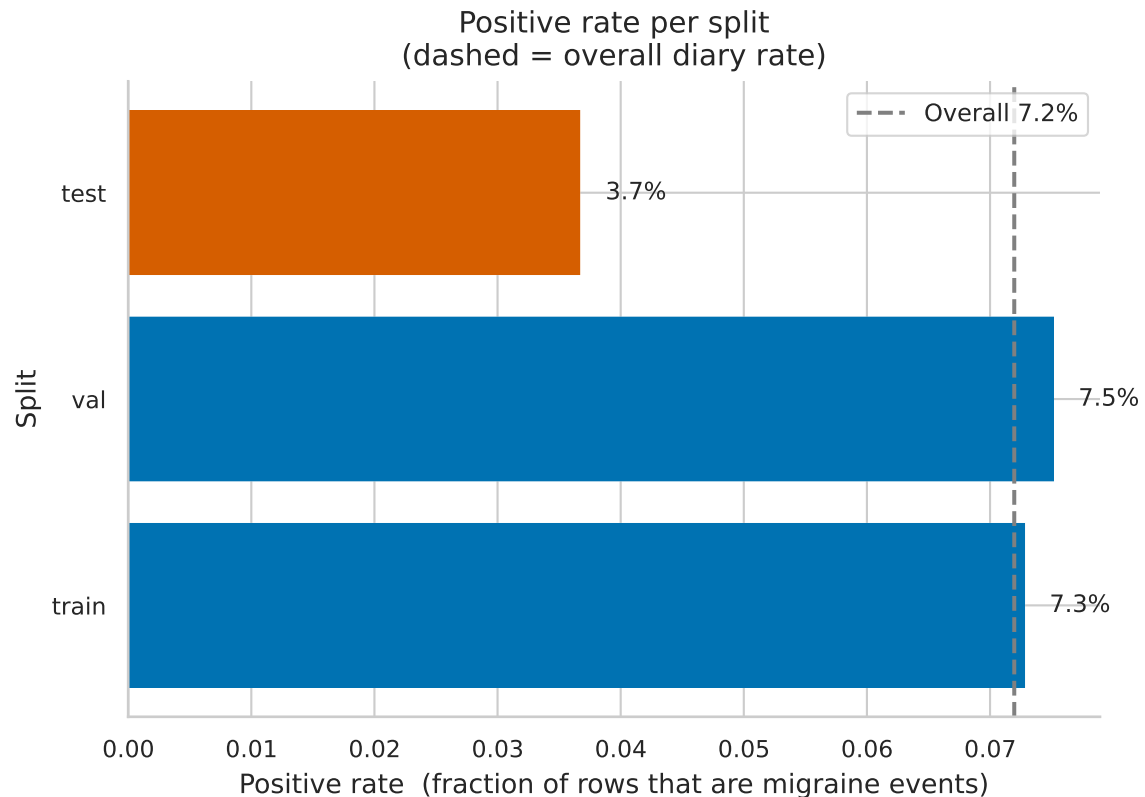
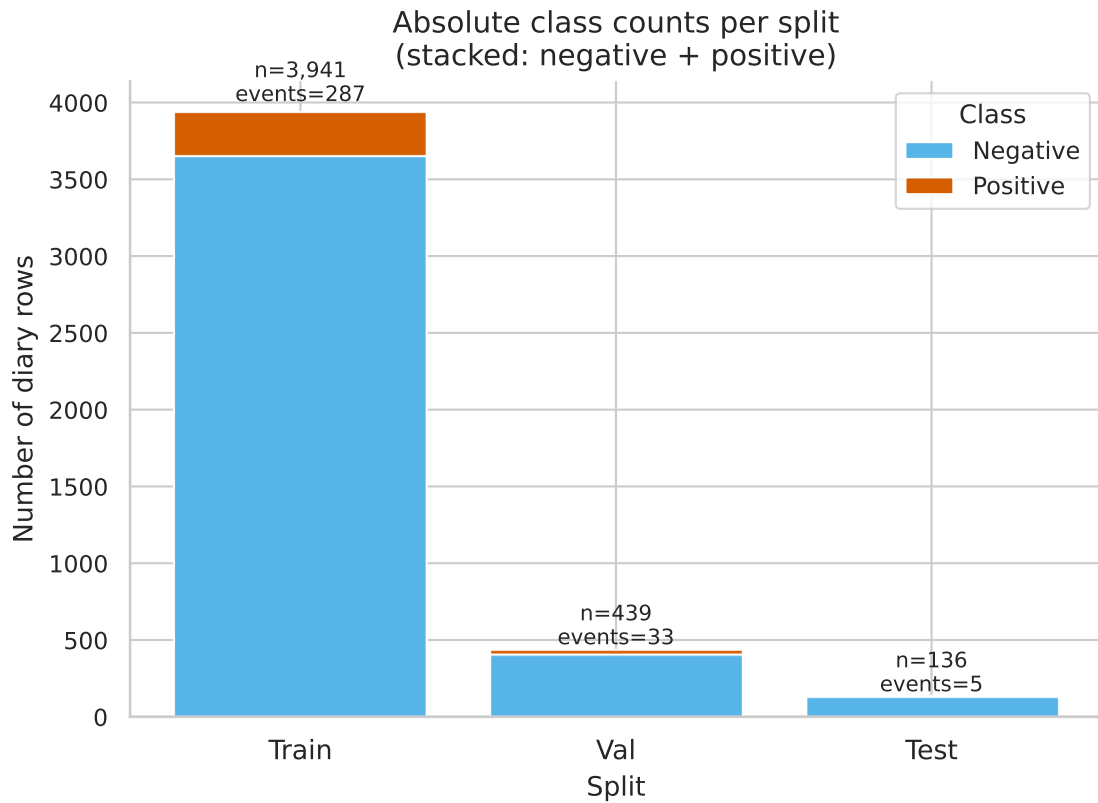


Class Balance Overview — migraine
(green: within 5pp of overall; yellow: 5-10pp; red: >10pp)

Package (ratio / strategy)	Train Rate	Val Rate	Test Rate	Train Events	Val Events	Test Events
70_15_15/chrono	7.3%	7.5%	3.7%	287	33	5
70_15_15/patient	7.0%	10.1%	5.5%	226	62	37
70_15_15/stratified	7.2%	7.2%	7.2%	227	49	49
80_20/chrono	7.4%	—	4.3%	315	—	10
80_20/patient	7.1%	—	7.5%	258	—	67
80_20/stratified	7.2%	—	7.2%	260	—	65
70_30/chrono	7.3%	—	6.6%	287	—	38
70_30/patient	7.0%	—	7.7%	226	—	99
70_30/stratified	7.2%	—	7.2%	227	—	98

Overall diary rate: 7.20%

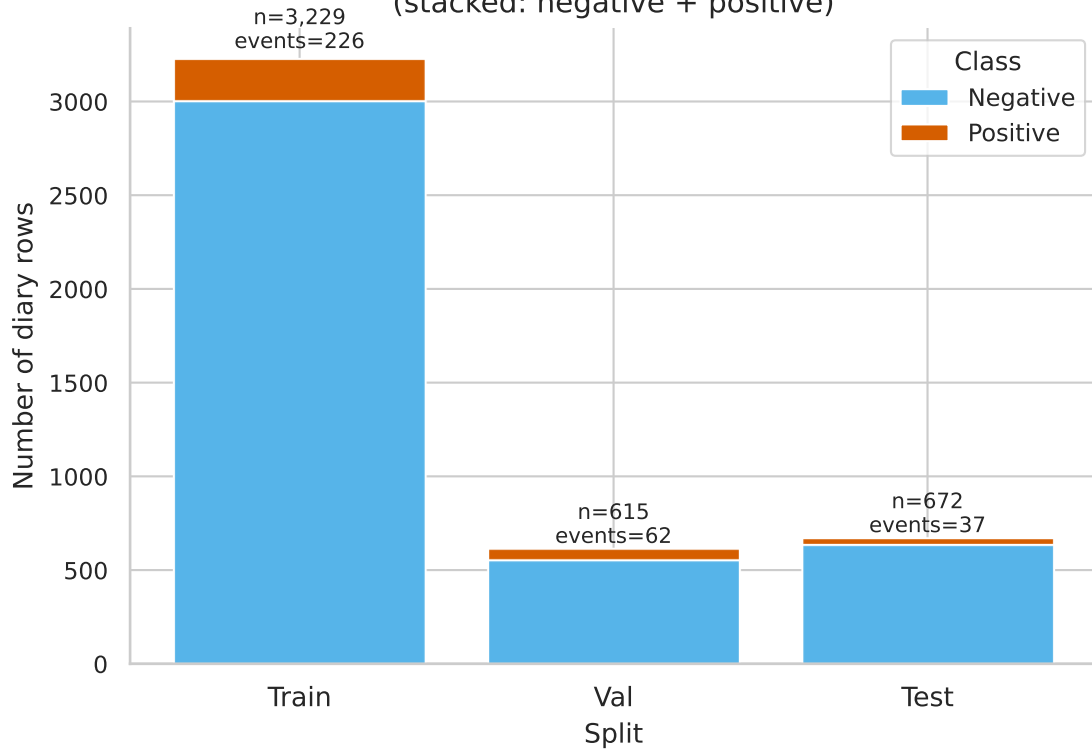
70_15_15/chrono — migraine



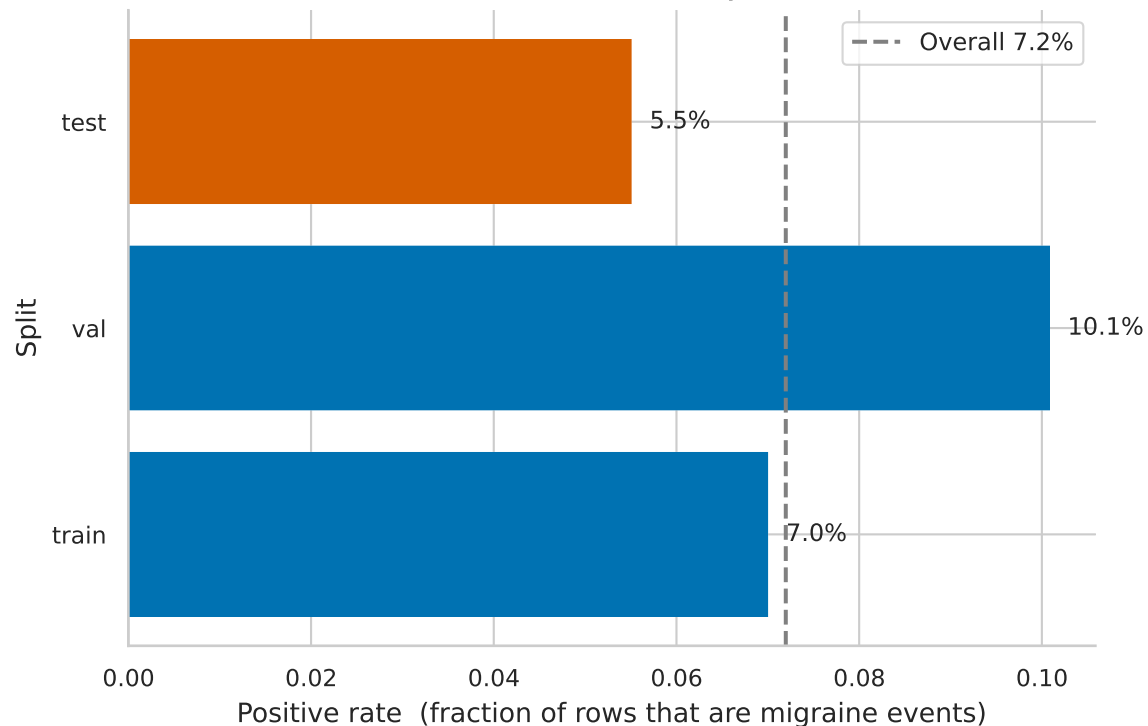
train: imbalance ratio 12.7:1 ($\Delta=+0.1\%$) | val: imbalance ratio 12.3:1 ($\Delta=+0.3\%$) | test: imbalance ratio 26.2:1 ($\Delta=-3.5\%$)

70_15_15/patient — migraine

Absolute class counts per split
(stacked: negative + positive)

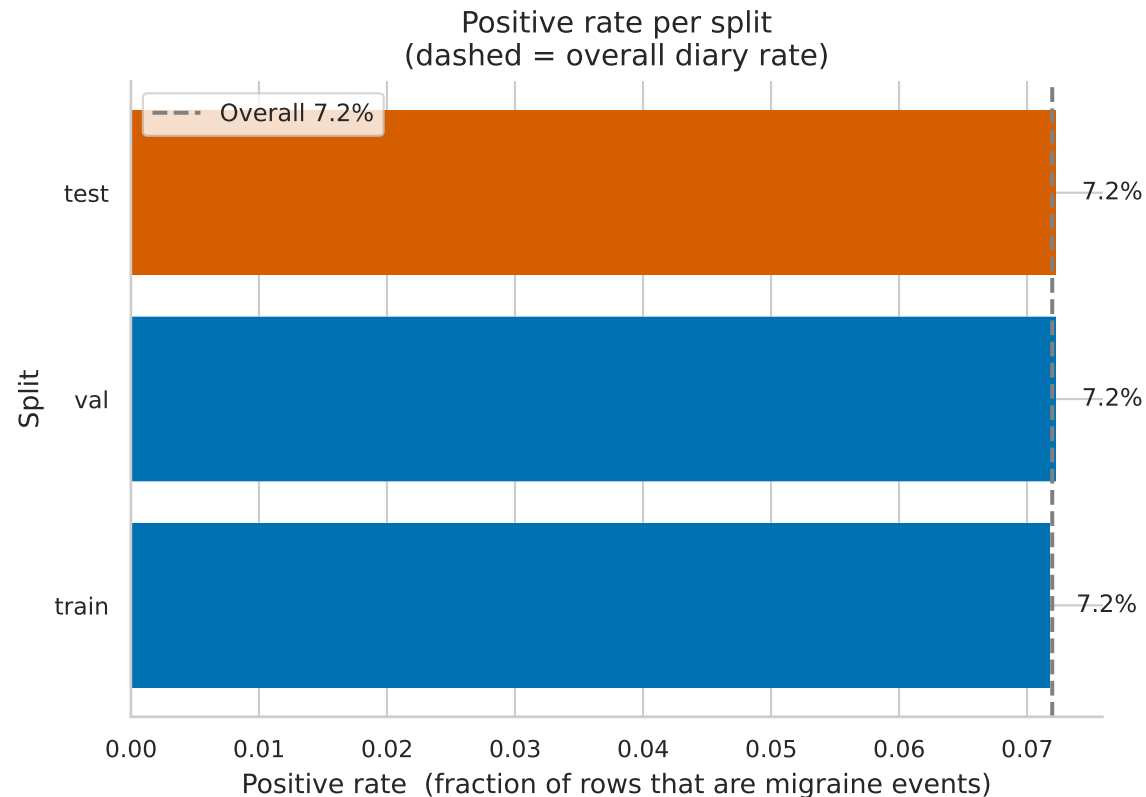
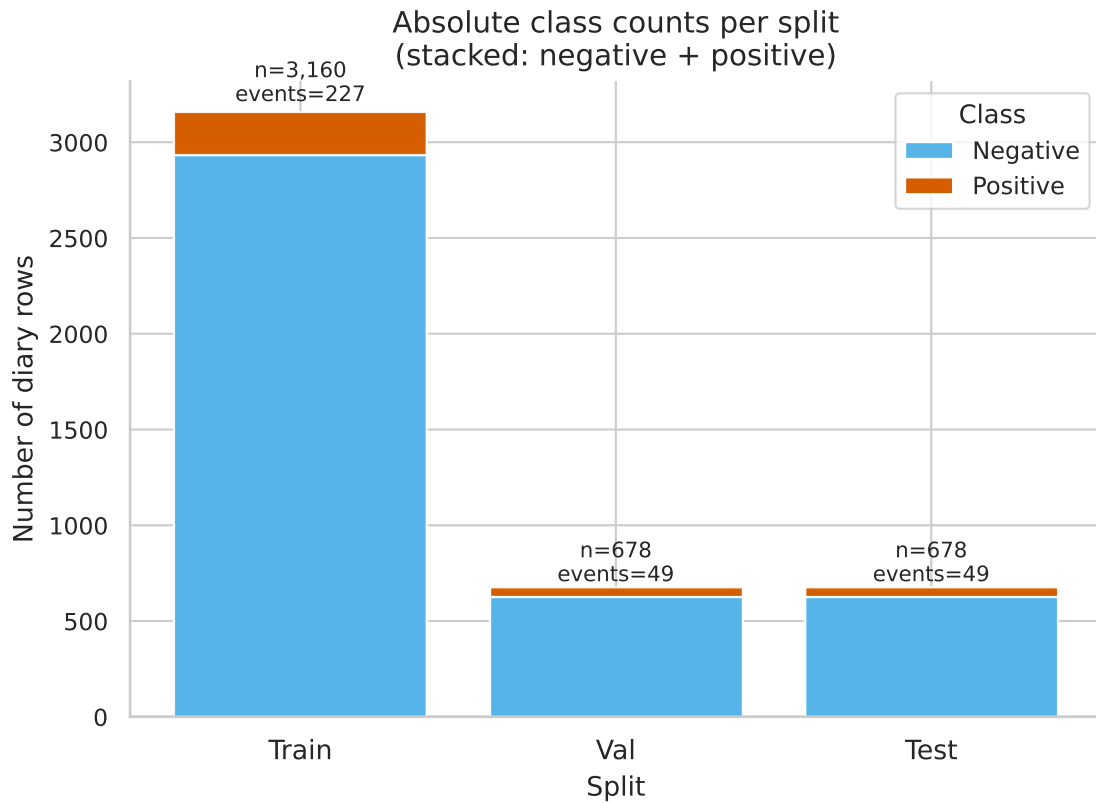


Positive rate per split
(dashed = overall diary rate)



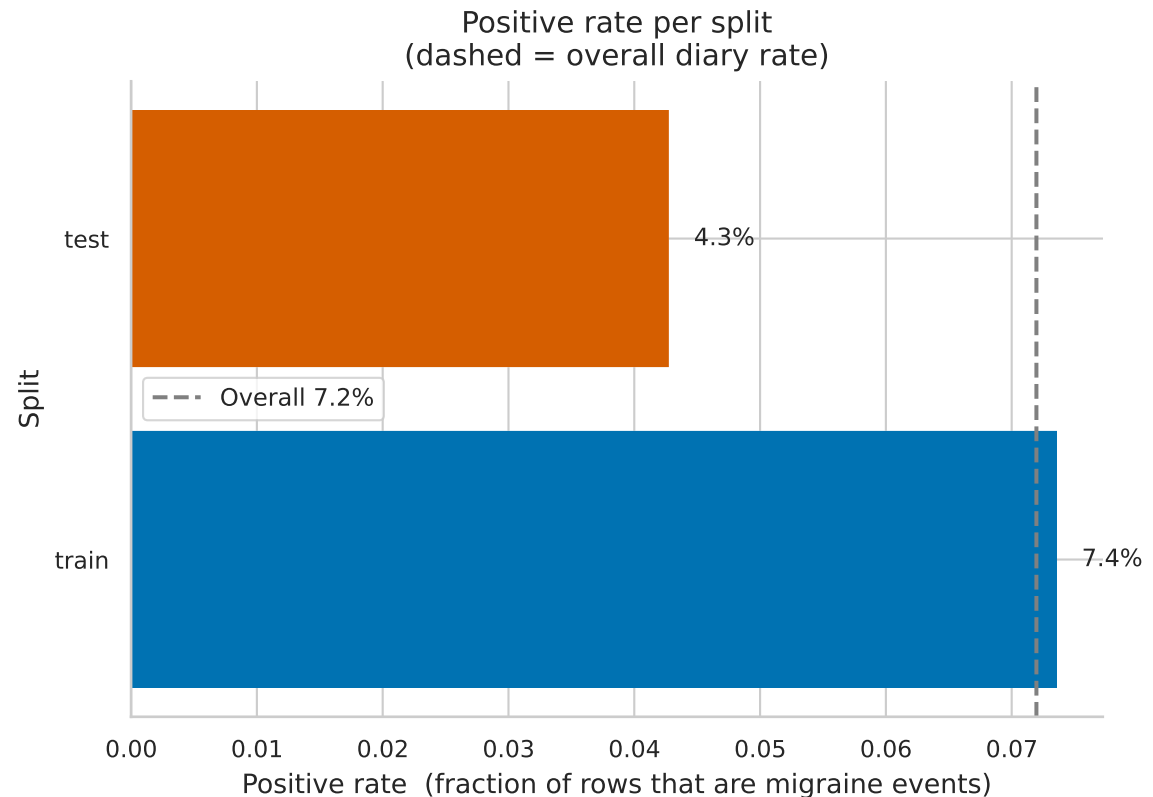
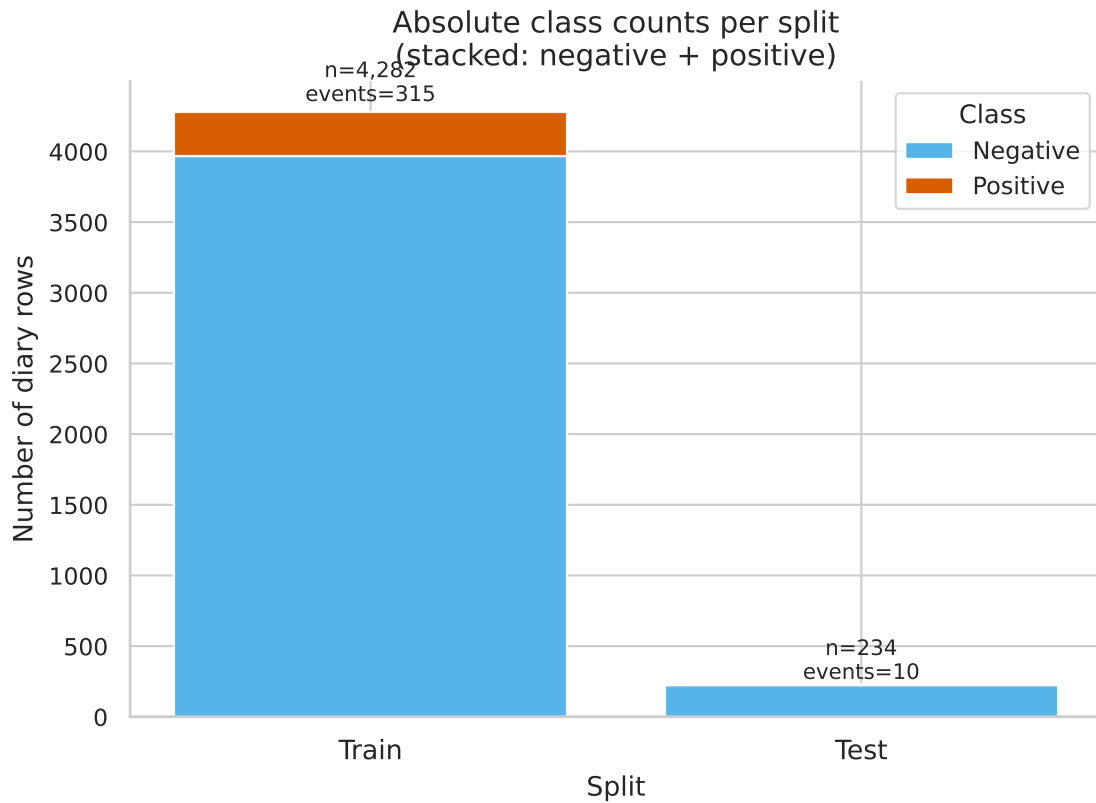
train: imbalance ratio 13.3:1 ($\Delta=-0.2\%$) | val: imbalance ratio 8.9:1 ($\Delta=+2.9\%$) | test: imbalance ratio 17.2:1 ($\Delta=-1.7\%$)

70_15_15/stratified — migraine



train: imbalance ratio 12.9:1 ($\Delta=-0.0\%$) | val: imbalance ratio 12.8:1 ($\Delta=+0.0\%$) | test: imbalance ratio 12.8:1 ($\Delta=+0.0\%$)

80_20/chrono — migraine

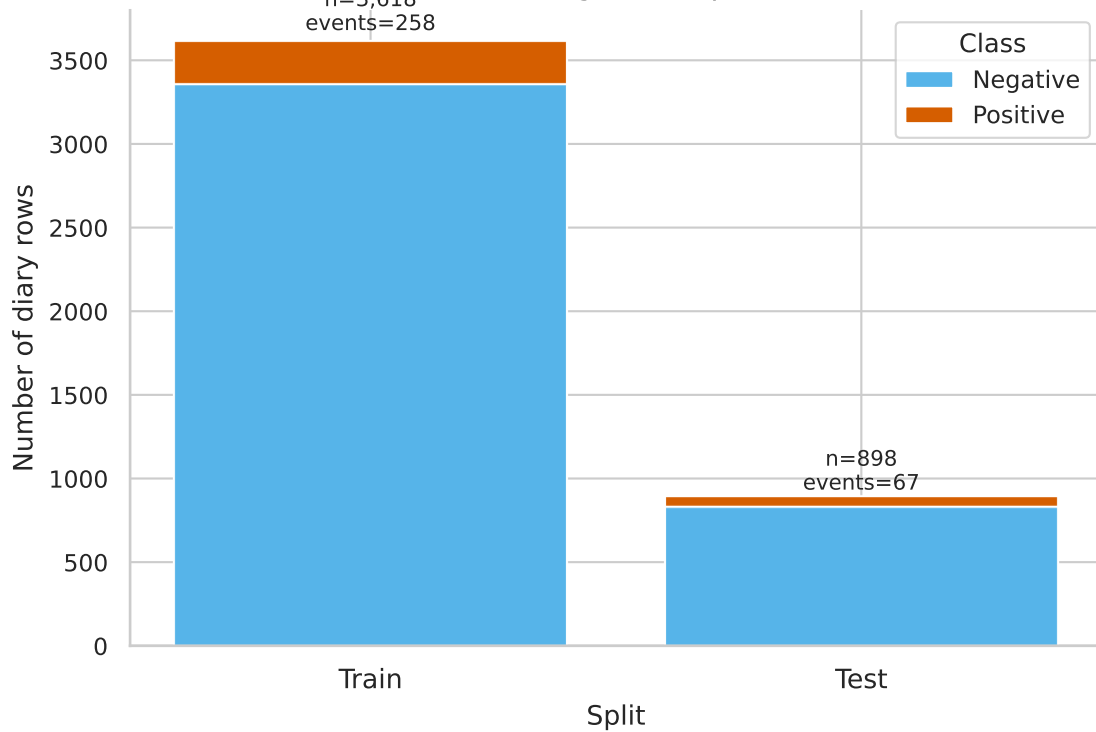


train: imbalance ratio 12.6:1 ($\Delta=+0.2\%$) | test: imbalance ratio 22.4:1 ($\Delta=-2.9\%$)

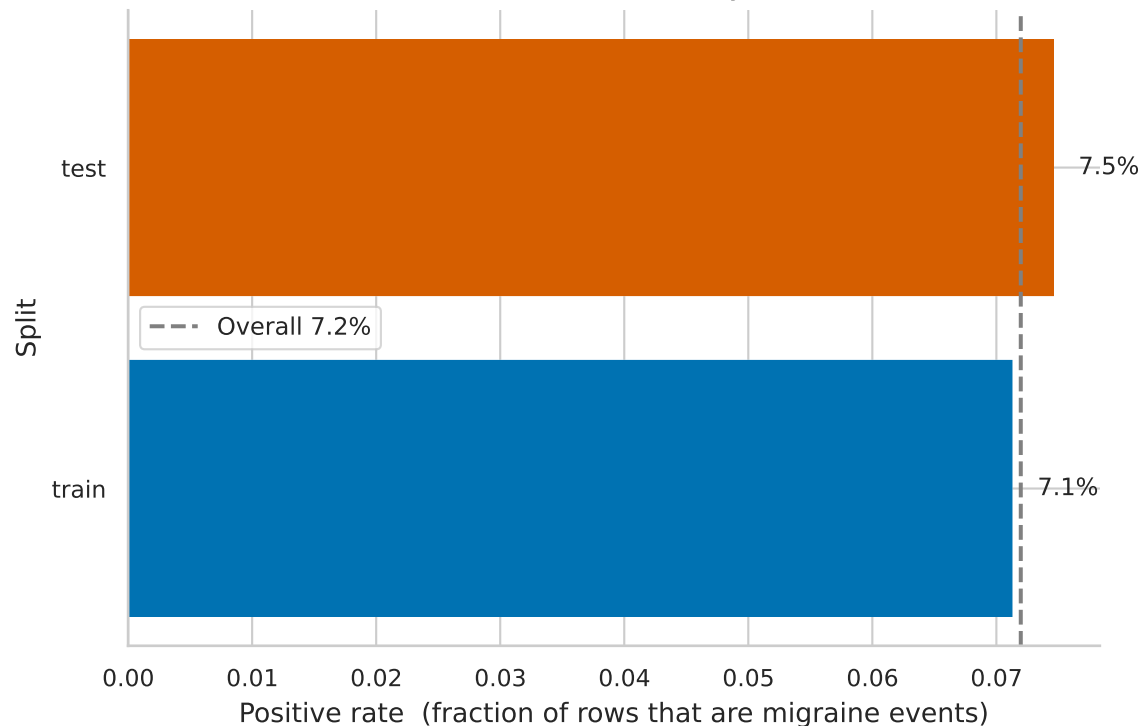
80_20/patient — migraine

Absolute class counts per split
(stacked: negative + positive)

n=3,618
events=258

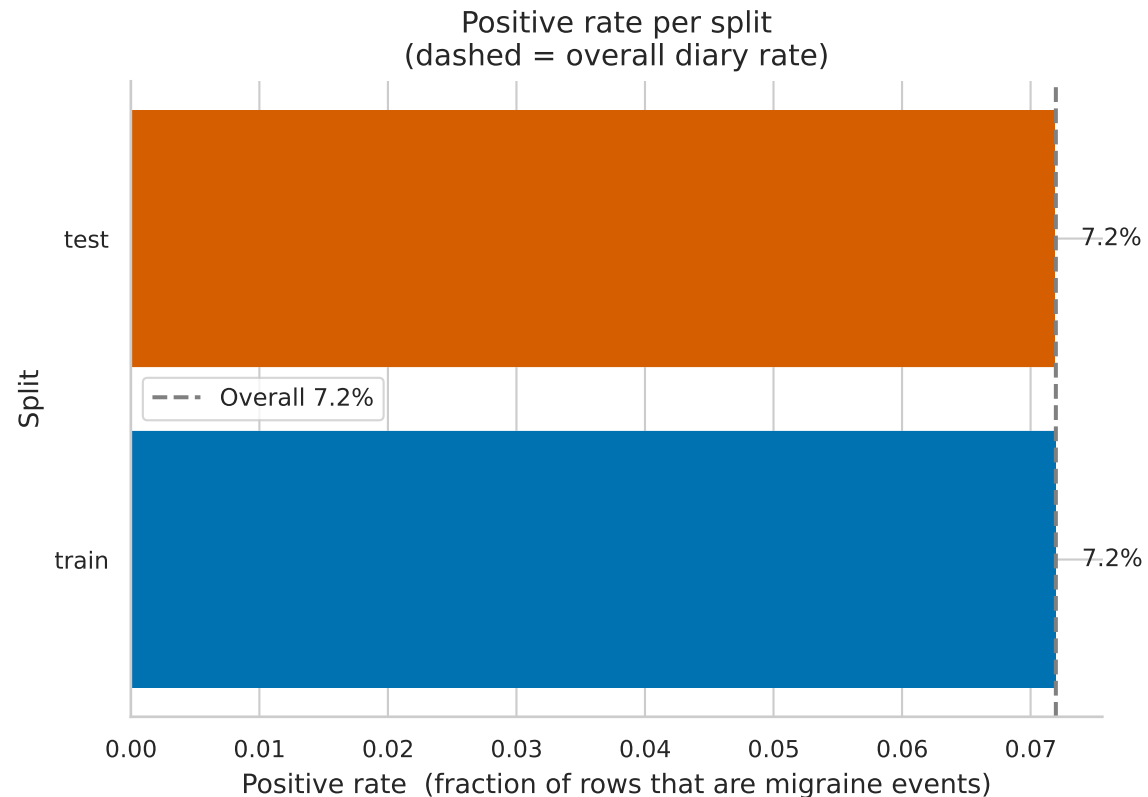
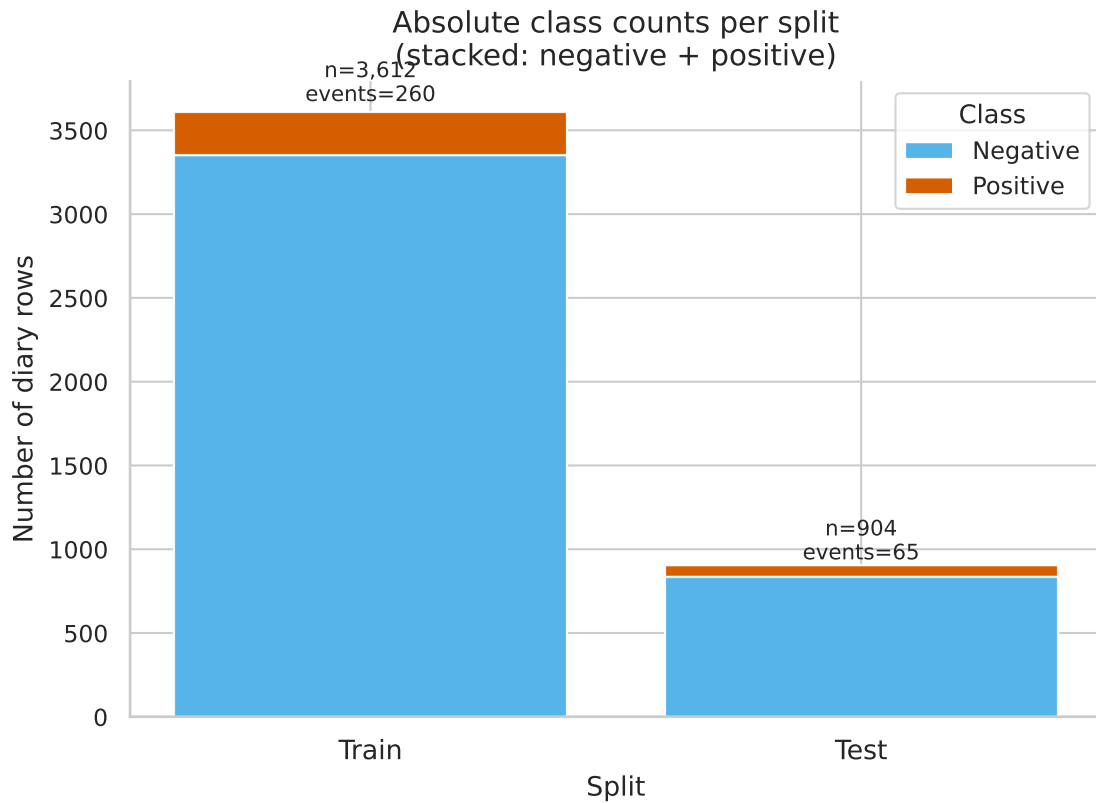


Positive rate per split
(dashed = overall diary rate)



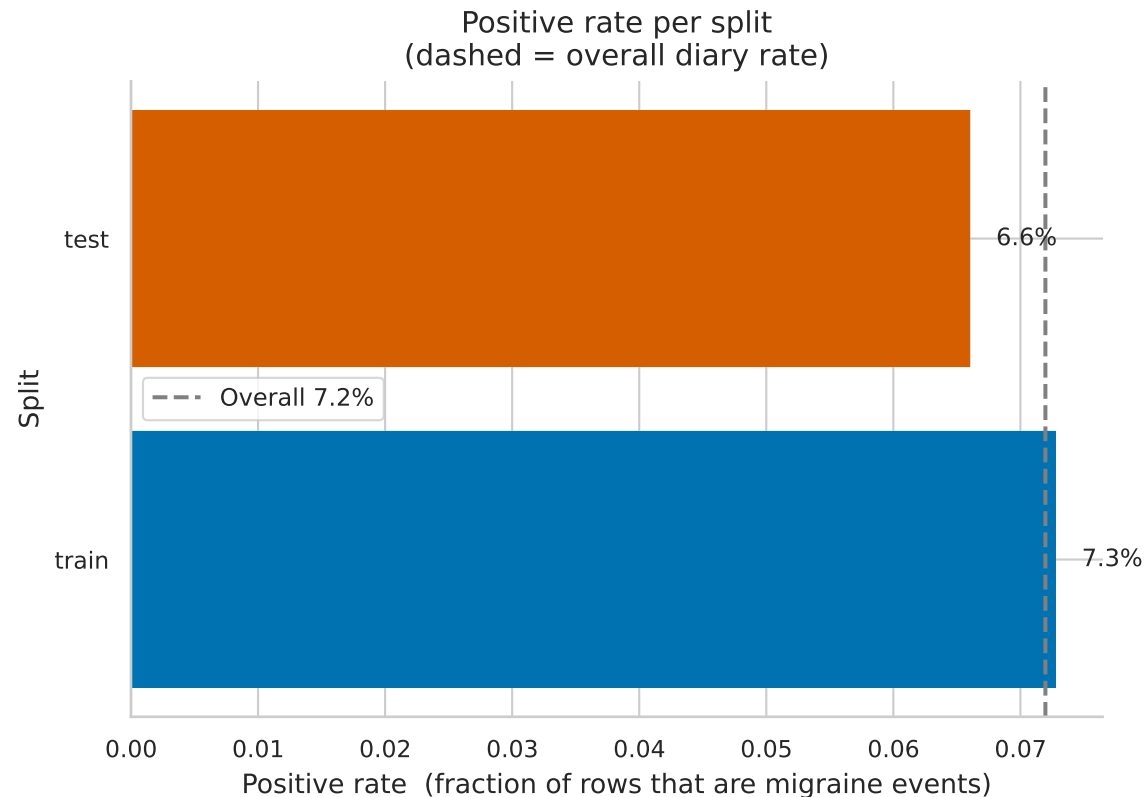
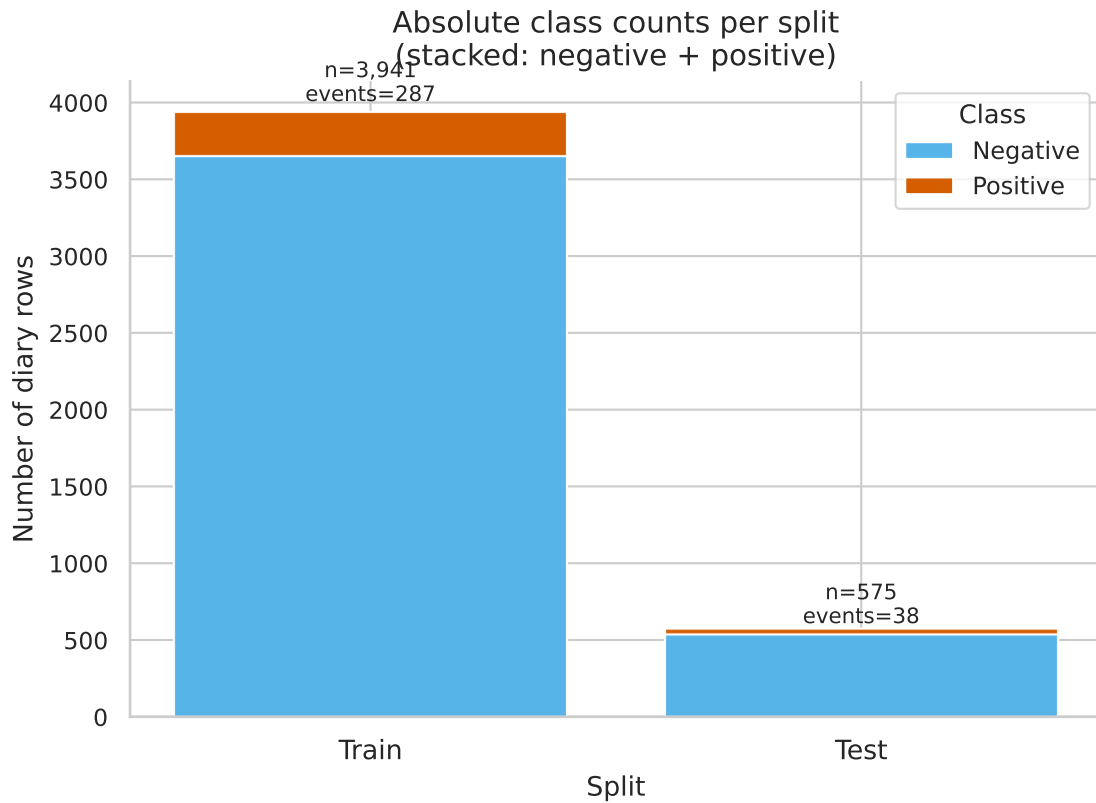
train: imbalance ratio 13.0:1 ($\Delta=-0.1\%$) | test: imbalance ratio 12.4:1 ($\Delta=+0.3\%$)

80_20/stratified — migraine



train: imbalance ratio 12.9:1 ($\Delta=+0.0\%$) | test: imbalance ratio 12.9:1 ($\Delta=-0.0\%$)

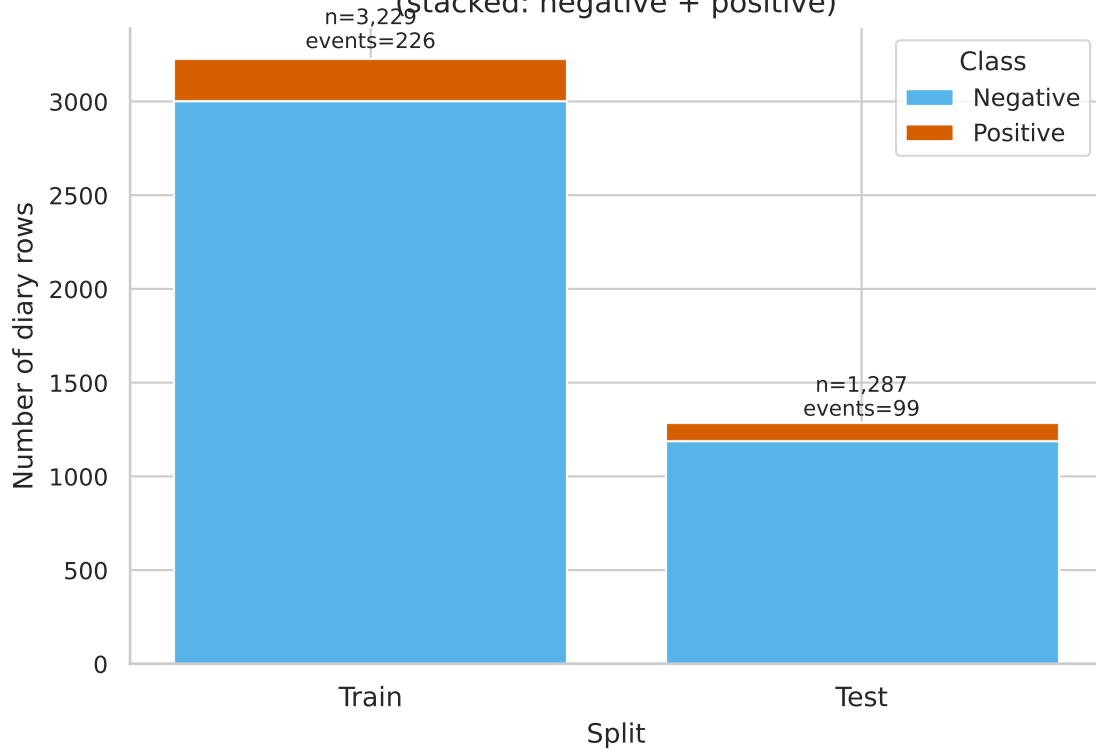
70_30/chrono — migraine



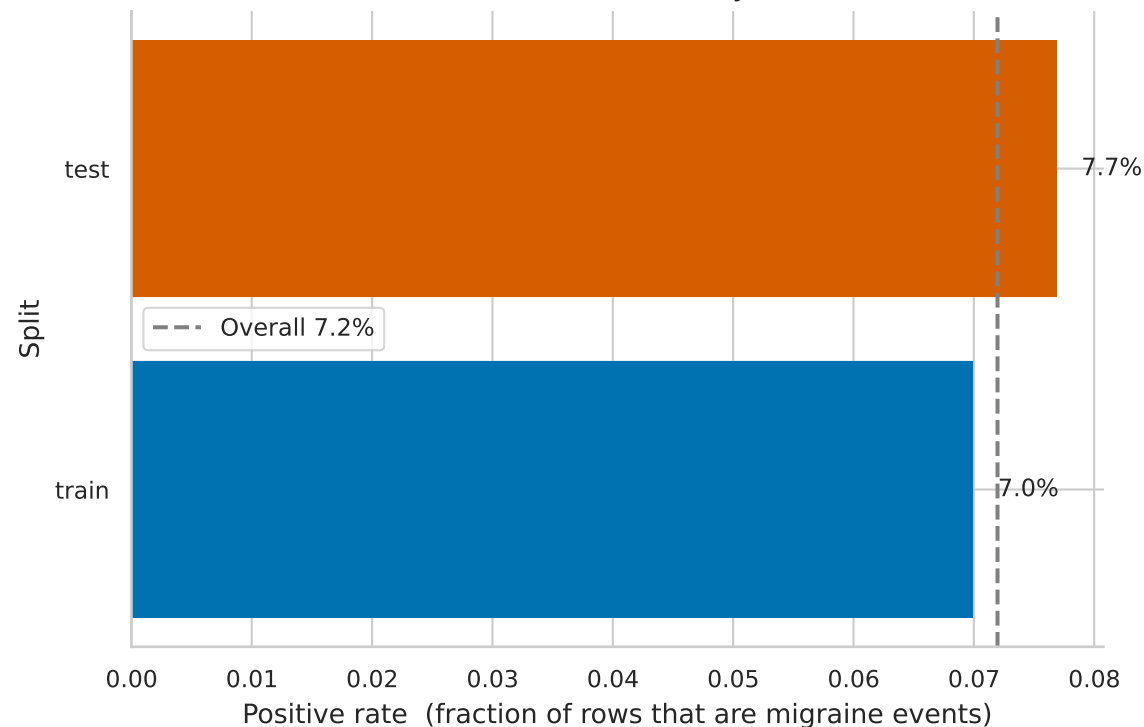
train: imbalance ratio 12.7:1 ($\Delta=+0.1\%$) | test: imbalance ratio 14.1:1 ($\Delta=-0.6\%$)

70_30/patient — migraine

Absolute class counts per split
(stacked: negative + positive)

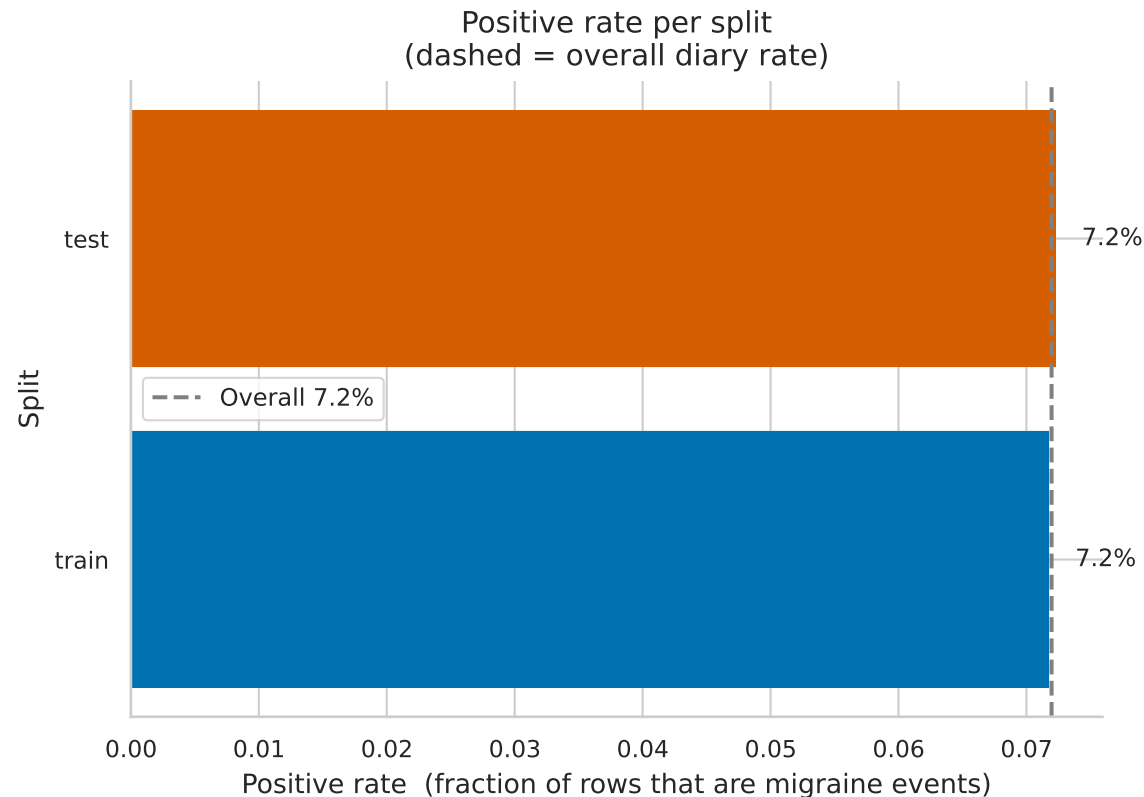
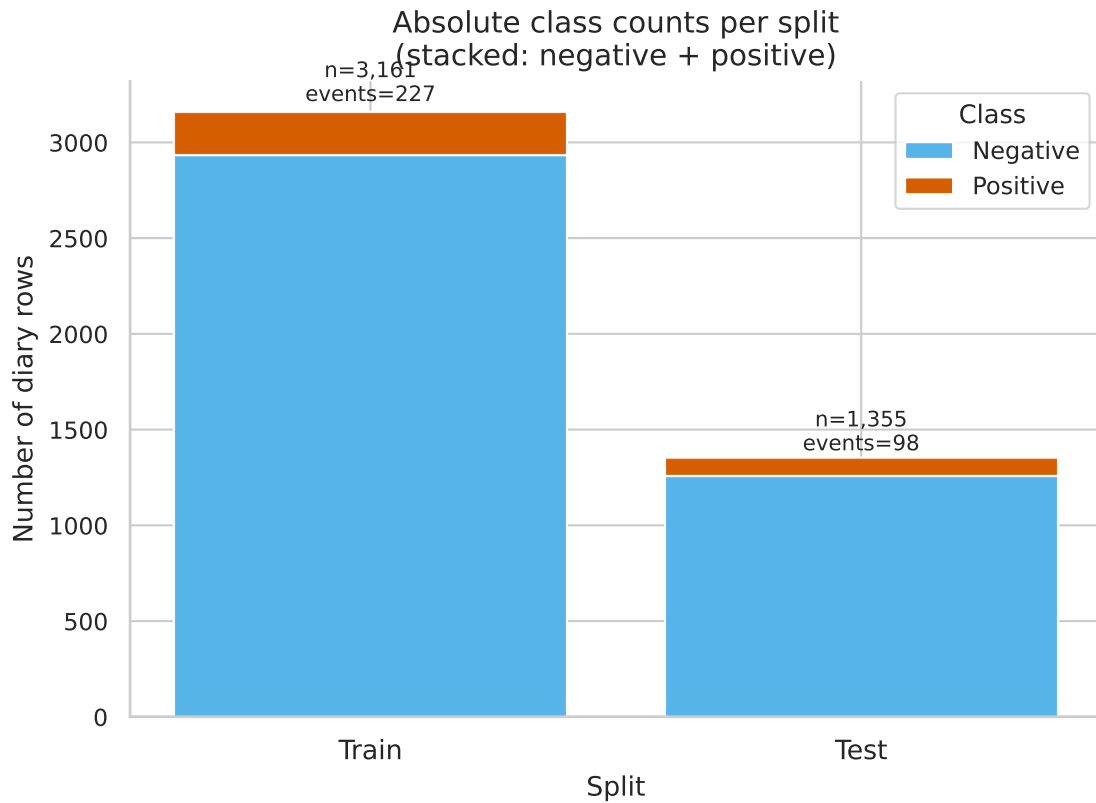


Positive rate per split
(dashed = overall diary rate)



train: imbalance ratio 13.3:1 ($\Delta=-0.2\%$) | test: imbalance ratio 12.0:1 ($\Delta=+0.5\%$)

70_30/stratified — migraine



train: imbalance ratio 12.9:1 ($\Delta=-0.0\%$) | test: imbalance ratio 12.8:1 ($\Delta=+0.0\%$)

Cross-package Test Positive Rate — migraine

Train vs test positive rate across all 9 packages
Red annotation = train-test delta > 5 percentage points

